

Logistic Regression

⇒ Decision Boundary:

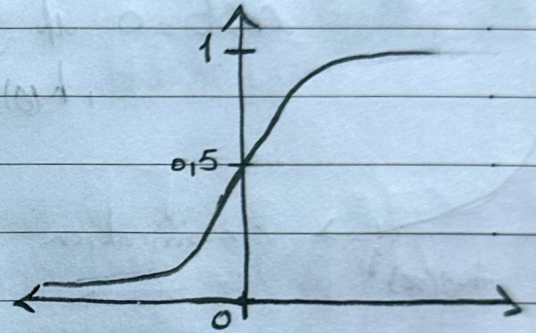
↳ $h_{\theta}(x) \geq 0$ considering squashing the line

* Squash → refers to the process of transforming a wide range of input values into a limited bounded range

$$\Rightarrow h_{\theta}(x) = g(\theta_0 + \theta_1 x)$$

$$\text{let } z = \theta_0 + \theta_1 x$$

$$h_{\theta}(x) = g(z)$$



↳ Sigmoid function :- "logistic Reg. func"

$$\Rightarrow h_{\theta}(x) = \frac{1}{1 + e^{-z}} \Rightarrow h_{\theta}(x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

Squashes line limits

⇒ ex - Training Set : $\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$
 $\Rightarrow y \rightarrow \{0, 1\}$ only 2 outputs

$$h_{\theta}(z) = \frac{1}{1 + e^{-z}}$$

$$\text{where } z = \theta_0 + \theta_1 x, \quad z(\theta, x)$$

← Logistic Reg - cost function :-

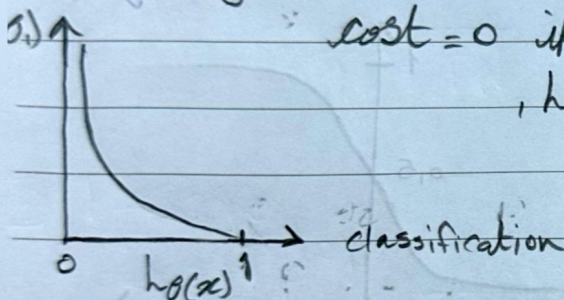
↳ solves the problem of non convexity

↳ cost func.

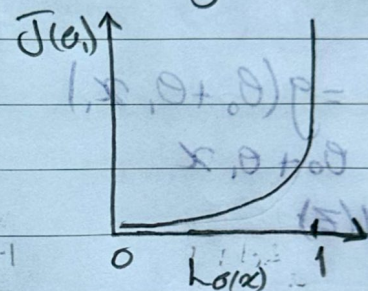
$$J(\theta) = \begin{cases} -\log(h_\theta(x^i)) & y=1 \\ -\log(1-h_\theta(x^i)) & y=0 \end{cases}$$

⇒ if $y=1$

cost = 0 if $y=1$, $h_\theta(x)=1$



⇒ if $y=0$



$$\Rightarrow \text{cost}(h_\theta(x^i), y) = \begin{cases} -\log(h_\theta(x)) & , y=1 \\ -\log(1-h_\theta(x)) & , y=0 \end{cases}$$

$$\Rightarrow h_\theta(x^i), y = -y \log(h_\theta(x^i)) - (1-y) \log(1-h_\theta(x^i))$$

↳ if $y=1$

$$\Rightarrow (h_\theta(x^i), y) = -\log(h_\theta(x^i))$$

↳ if $y=0$

$$\Rightarrow (h_\theta(x^i), y) = -\log(1-h_\theta(x^i))$$

* final cost func formula :-

$$J(\theta) = -\frac{1}{2m} \sum_{i=1}^m (y^{(i)} \log(h_{\theta}(x^{(i)})) + (1-y^{(i)}) \log(1-h_{\theta}(x^{(i)})))$$

$$\text{where: } h_{\theta}(x^{(i)}) = \frac{1}{1 + e^{-(\theta \cdot x)}}$$

2 → convergence algorithm for Logistic Reg.

$$\theta_j = \theta_j - \alpha \frac{\partial (J(\theta))}{\partial \theta_j}$$

* Performance Matrix "classification problem"

Actual \ Predict		0	1
Predict	0	3	2
	1	1	1

Confusion matrix

TP	FP
FN	TN

TP = true + Ve

TF = true - Ve

FP = false + Ve

FN = false - Ve

$$* \text{accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{3 + 1}{3 + 2 + 1 + 1} = \frac{4}{7} = 57\% \text{ accuracy}$$

* our aim to reduce (FP, FN)

Notes

لنقاش لازم اعتمد على accuracy لانهم يمكن ان يكون Bias data
 كـي يعني انهم مثلاً قديم $900 \rightarrow 100$ $1000 \rightarrow 1$
 كـالها أمرب الموديل الـ acc 100% بس غلط

الحلول

$$1) \text{ Precision} = \frac{TP}{TP + FP}$$

used in spam classification

$$2) \text{ Recall} = \frac{TP}{TP + FN} \rightarrow \text{Heart disease classification}$$

$$3) \text{ F-Score or F-}\beta$$

$$\rightarrow (1 + \beta^2) \cdot \frac{\text{Precision} \cdot \text{Recall}}{\beta^2 (\text{Precision} + \text{Recall})} = \frac{2 (\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}}$$

Harmony mean $\frac{2xy}{x+y}$

here the focus is on FP and FN ; FP is more important than FN that decreases β value

$\Rightarrow$ decreasing β value $\rightarrow$ Providing more importance to FP than FN

so it's done according to what's important to me